BOAT PROJET

FIRST SEMESTER

TEAM 1



MEMBERS :

NAME	Domain
NYA'ABE NDO ANDY RAPHAELLE	MEO
NJOUGUEP WANDJI EMMANUELLE	ESE
ABO WILLIAM JUNIOR	EEE, MIA
LOUNDOU-LOUNDOU Ezechiel	MME MAT - MME MEC
MAKOSSO Mignon	MIA

Actor and Referent

Actor	Referent
Project manager	NYA'ABE NDO ANDY RAPHAELLE
Referent boat	MOUANDA Maxence

GENERAL SUMMARY

- INTRODUCTION
- STUDY PLAN
- CONCLUSION

GENERAL STUDY PLAN

1

MEO

Learn to manage deliverable in a multidisciplinary project

2

ESE

Studies on boat stability

3

MME MAT

4

MME MEC

5

EEE

Design, simulate and test motor control system blocks using appreciate software and hardware

6

MIA

Design and construction of an autopilot

INTRODUCTION

Throughout these months, we have been asked to theoretically design a boat hull according to our preferences and respects well-established standards. It is therefore with this in mind that we present this summary to you

MEO DOMAIN

1. Project Presentation
2. Objectives and challenges
3. SWOT Method
4. Project Planning
5. Requirement analysis
6. Cost of the project

Animator: NYA'ABE NDO Andy Raphaelle

1- Project Presentation

In this first semester, our team had the task of designing a one-piece boat measuring 70 cm in length meeting the defined constraints.
Schedule: for this first phase we have a duration of 93 hours or 13 weeks broken down as follows.

Hours	Distribution
54 Hours	Group work by domain
39 Hours	empowered

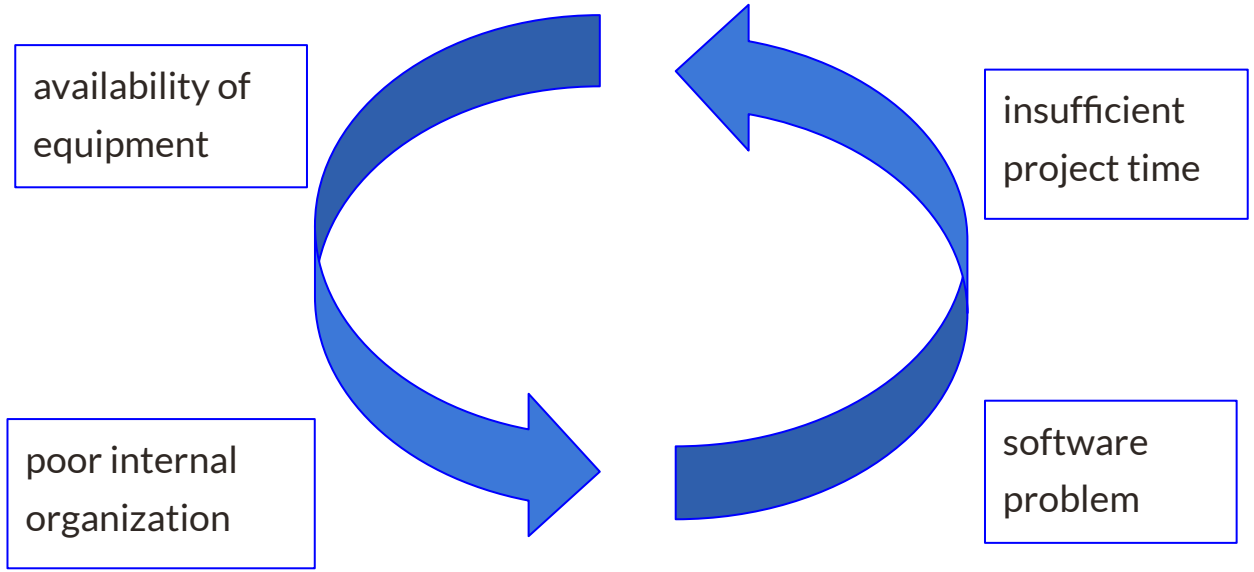
2- Objectives and challenges

OBJECTIVES

Domaines	Objectives
MEO	Be able to manage the project as well as the various deliverables while maintaining a climate of cohesion between the members of the team.
ESE	Produce a report about the stability and the floatability of the boat.

MME MEC	Carry out the first dimensioning and design the hull with suitable digital tools .
MME MAT	Choose the right material for the hull of our boat while taking into account the environment to which it will be subjected.
EEE	Break down the motor control system into functional blocks, design and simulate the functional model.
MIA	Design and build an autopilot experimentally.

CHALLENGES


availability of equipment
insufficient project time
poor internal organization
software problem

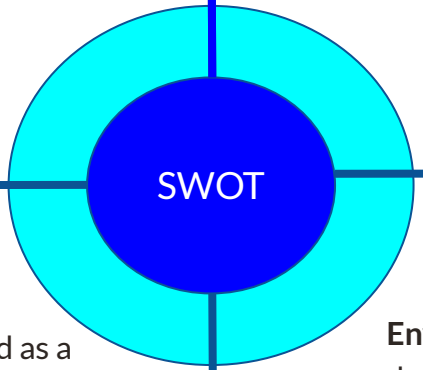
3- SWOT

STRENGTHS

Lightweight material : the emphasis on lightweight materials will optimize the buoyancy of the product
Ecological focus: the use of electric energy goes hand in hand with sustainability trends
Acquisition of skills: this project allows the development of technical skills

WEAKNESS

Absence of practical test: the theoretical study in the first year does not provide real technical validation
Limited experience: First nautical experience



OPPORTUNITIES

Documentation: the project can be used as a reference for future work or to be included in a portfolio professional.
Skills Building: an Experience unique interdisciplinary design, management, and prototyping, which prepares you for the professional world

THREATS

Environmental factors: Unpredictable conditions during tests (wind, waves, water quality).
Time pressure: Although over two years, the distribution in semesters makes each phase critical to overall success.

4- PROJECT PLANNING

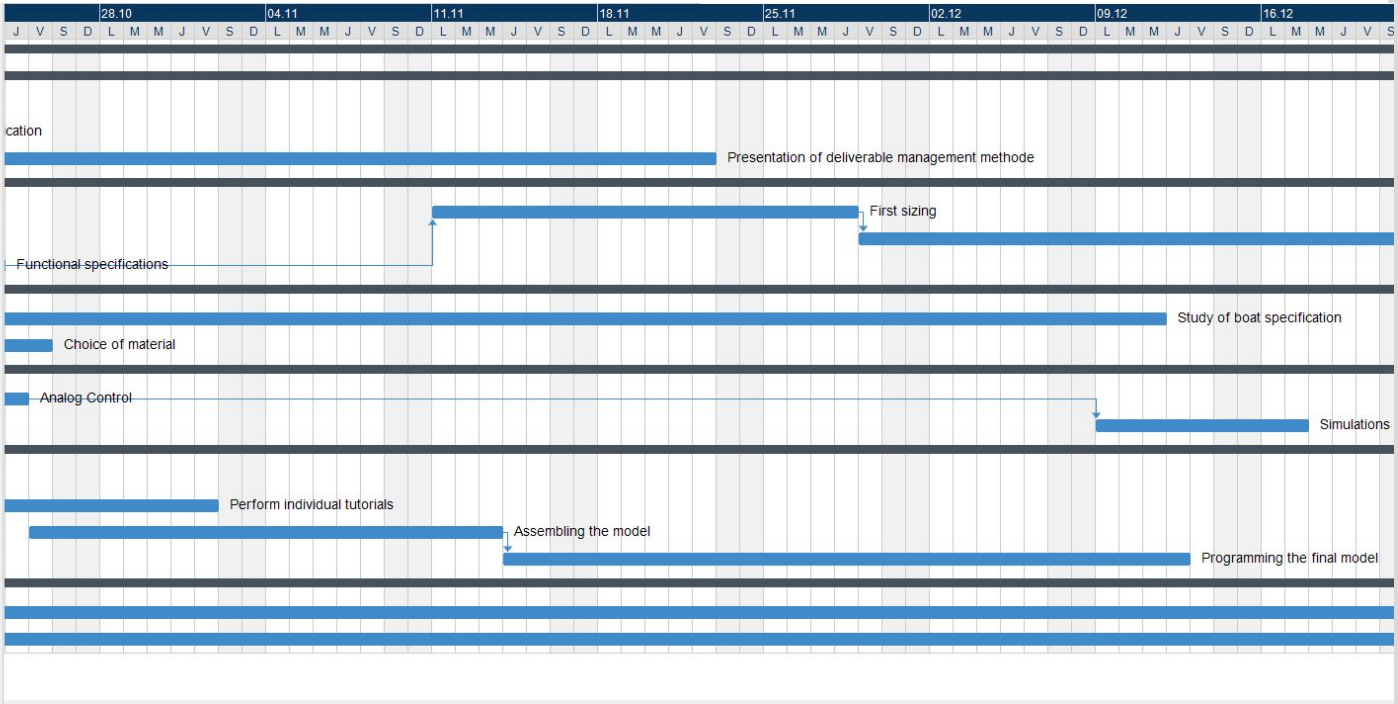
Our project was planned using two diagrams:

The Gantt chart: The Gantt chart, commonly used in project management, is one of the most effective tools for visually representing the progress of the different activities (tasks) that constitute a project

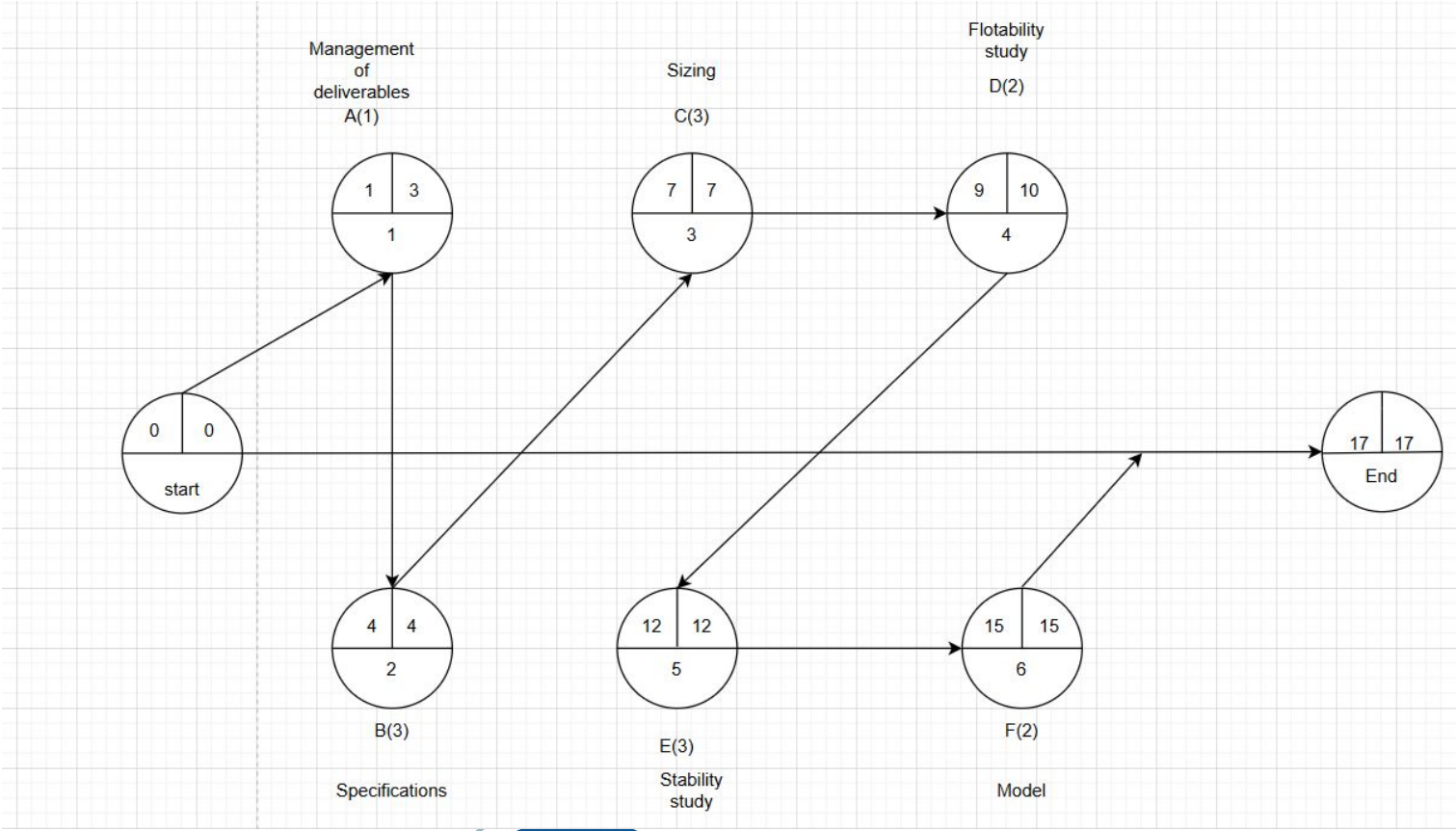
The PERT diagram: allows you to identify connections between different tasks, execution times, interdependencies.

GANTT Chart

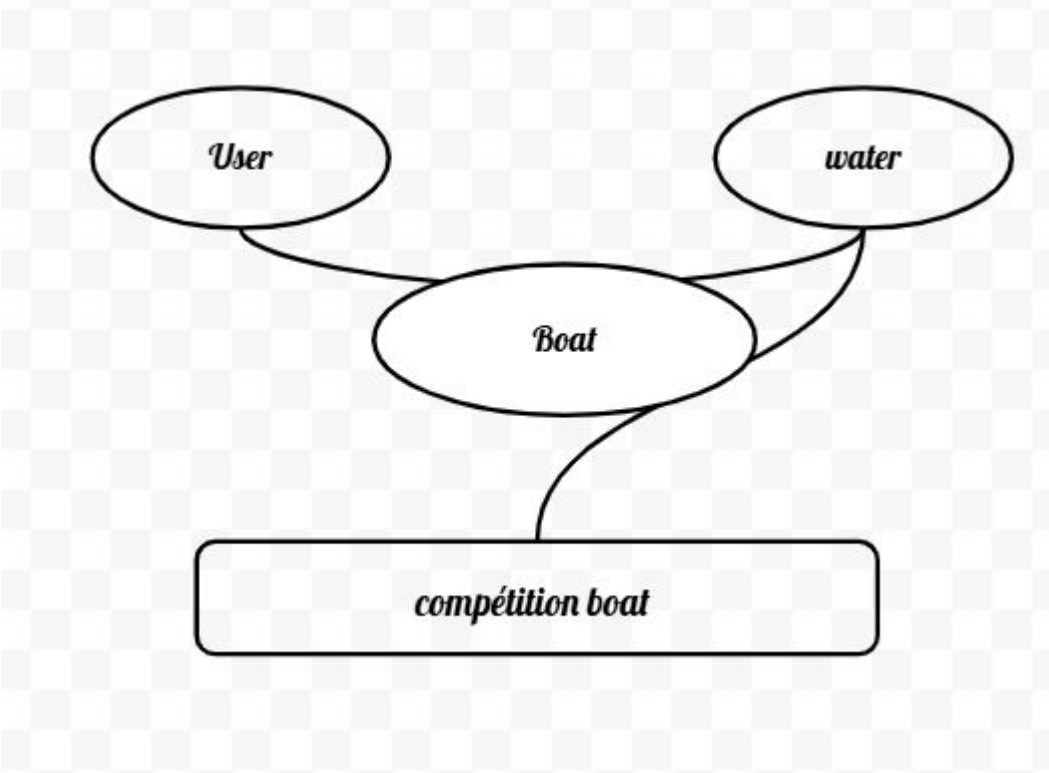
Pla…	Nom de tâche	Début	Fin
	BOAT PROJECT	**02/09/2024**	**27/01/2025**
	MEO Domain	**02/09/2024**	**27/01/2025**
	Project ani…	02/09/2024	23/09/2024
	Manageme…	24/09/2024	16/10/2024
	Presentatio…	17/10/2024	22/11/2024
	MME MEC…	**02/09/2024**	**27/01/2025**
	First sizing	11/11/2024	28/11/2024
	CAO pres…	29/11/2024	27/01/2025
	Functiona…	05/09/2024	23/10/2024
	MME MAT	**02/09/2024**	**15/01/2025**
	Study o…	04/10/2024	11/12/2024
	Choice …	16/09/2024	25/10/2024
	EEE D…	**02/09/2024**	**15/01/2025**
	Analo…	07/10/2024	24/10/2024
	Simul…	09/12/2024	17/12/2024
	MIA…	**02/09/2024**	**15/01/2025**
	Proj…	04/10/2024	14/10/2024
	Perf…	15/10/2024	01/11/2024
	Ass…	25/10/2024	13/11/2024
	Pro…	14/11/2024	12/12/2024
	ES…	**02/09/2024**	**15/01/2025**
	S…	02/09/2024	15/01/2025
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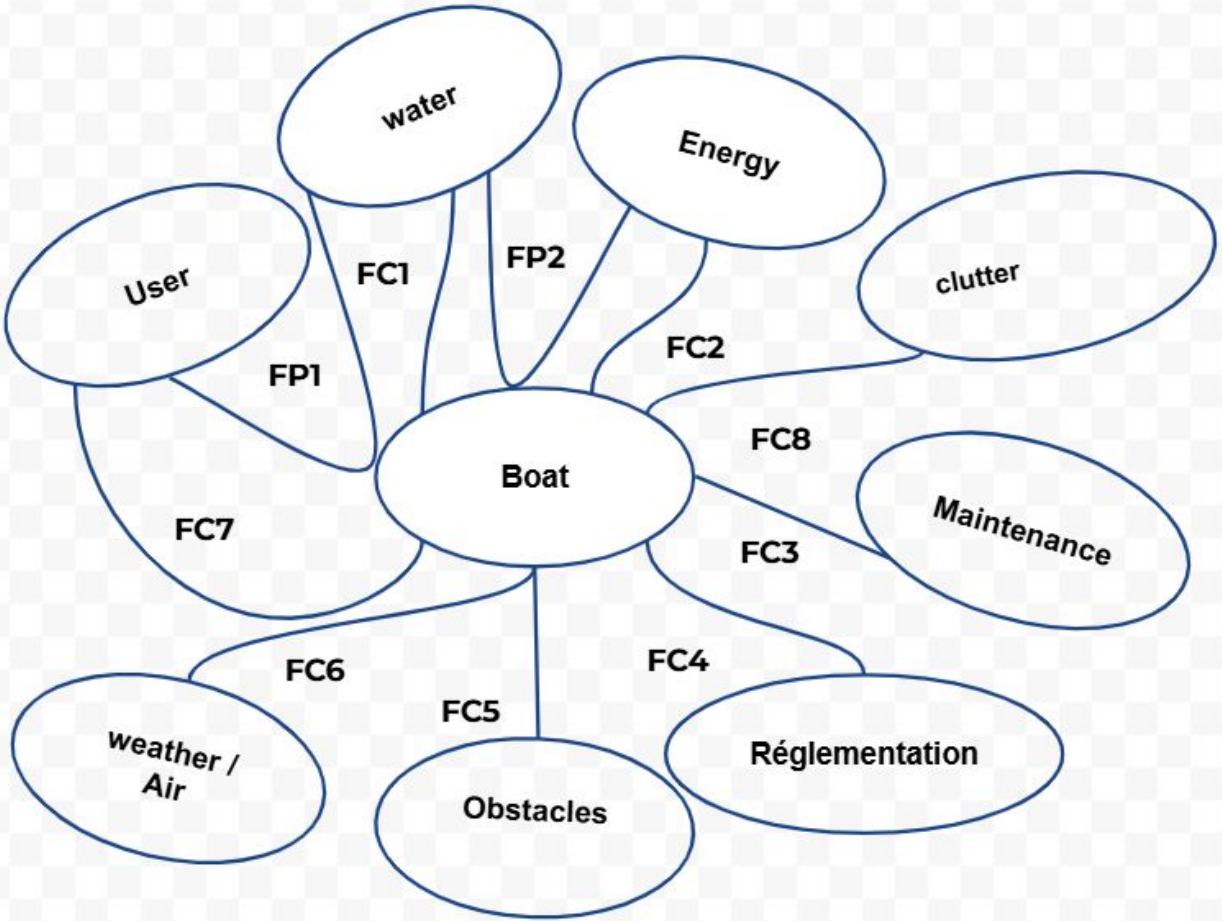


PERT DIAGRAM



5- Requirement analysis




water
Energy
User
FC1
FP2
clutter
FC2
FP1
FC8
Boat
Maintenance
FC7
FC3
FC6
FC4
FC5
weather /
Air
Réglementation
Obstacles

Glossary :

FP1: Move on the water following the user's commands
FP2: Allows you to propel the boat
FC1: Float and move on the water without capsizing
FC2: Have an autonomous energy source to power the propulsion and the control system
FC3: Be easy to maintain to prolong its durability
FC4: Comply with regulations relating to remote-controlled devices
FC5: Capable of detecting and avoiding obstacles and unidentified floating objects (OFNI)
FC6: Navigate in varying weather conditions
FC7: Have an easy-to-use control system
FC8: Present reduced dimensions for ease of use

6- costs of the project

An arduino microcontroller board	49,99€(Amazon)
A breadboard	2,422 XAF (Ali express)
A DC motor driver	2376,58 XAF (Micro planet)
A potentiomètre	129.89 $ (Amazon.fr)
An LCD Screen	XAF 3664.74 (Jumia)

for our project the estimation of the cost is 39173.64 XAF

MME-MEC DOMAIN

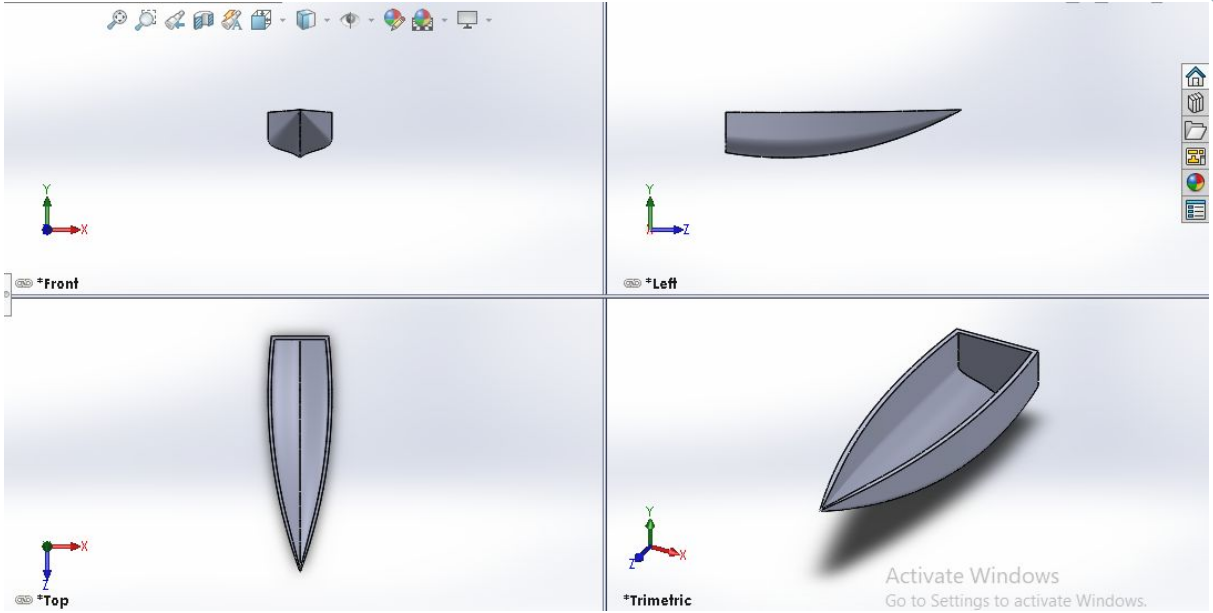
Function	FP1	FP2	FC1	FC2	FC3	FC4	FC5	FC6	FC7	FC8
Designation	Enables movement	Enables propulsion	Must float and remain stable	Must have a reliable energy source	Must resist corrosion	Must comply with regulations	Must detect and avoid obstacles	Must navigate in various weather conditions	Must have an easy-to-use control system	Must navigate even when obstructed
Criterion	Mass	Buoyancy force	Weight and Archimedes' force	Autonomy	Corrosion rate	Compliance with ISO standards	Maneuverability	Righting	Simplified operation system	Capacity to maneuver with bulky objects
Level	> 5 Kg	Fp > water resistance	Buoyancy/weight = 1	2 hours	Stainless	Mandatory	Maneuverable	Heel angle ≥ 25°	Maneuverable	High
Flexibility	F2	F0	F0	F2	F0	F0	F1	F0	F1	F1

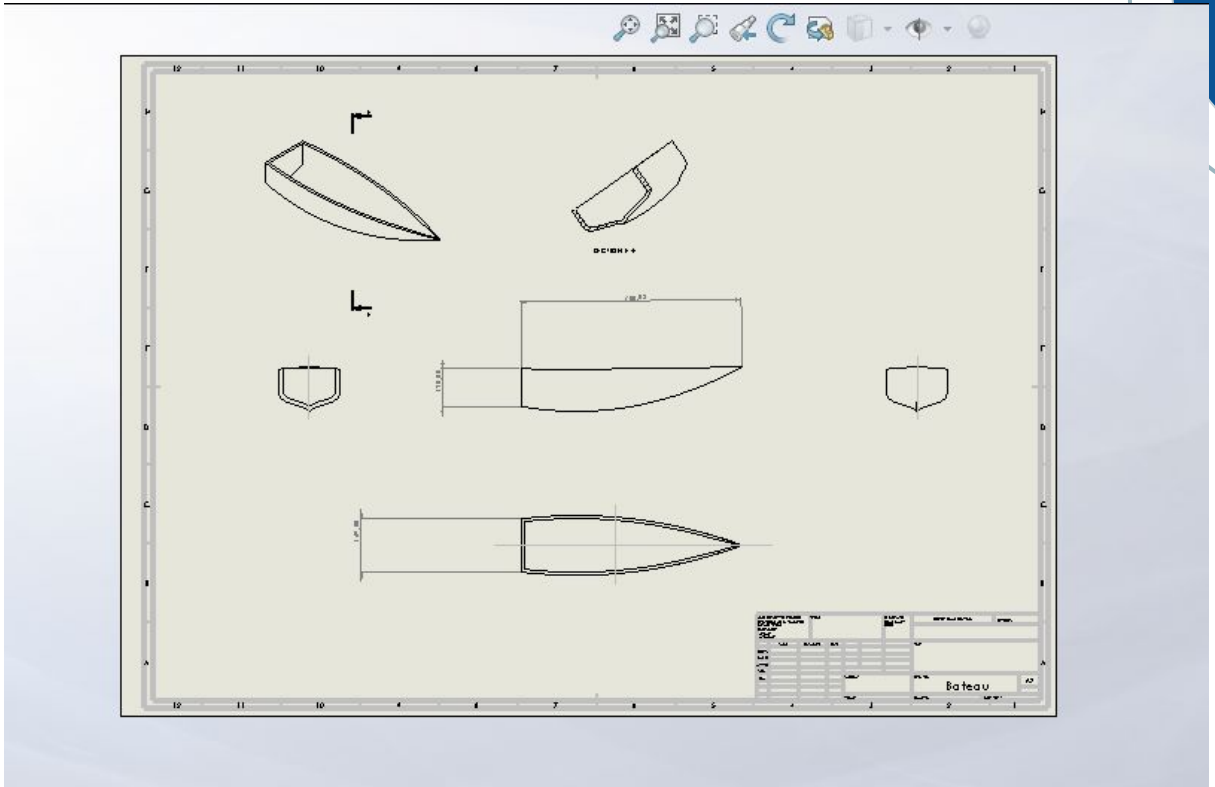
Sizing plan

Quantity	Value
Length (L)	70 cm
Load (Mass)	532 g
Beam (B)	16.98 cm
Draft (D)	8 cm
Freeboard (F)	7 cm
Waterline (T)	5 cm
Height (H)	7.5 cm

Parameter	Formula/Condition	Result
Width B	$B = (L)^{2/3}$, where $L = 70$ cm	$B = 16.98$ cm
Depth D	$1.55 \leq \frac{B}{D} \leq 2.5$ or $\frac{B}{2.5} \leq D \leq \frac{B}{1.55}$	$6.792 < D < 10.95$ cm, chosen $D = 8$ cm
Freeboard F	$1.55 \leq \frac{B}{F} \leq 2.5$ or $\frac{B}{2.5} \leq F \leq \frac{B}{1.55}$	$6.792 < F < 10.95$ cm, chosen $F = 7$ cm
Water Draft T	$2.5 \leq \frac{B}{T} \leq 3.75$ or $\frac{B}{3.75} \leq T \leq \frac{B}{2.5}$	$4.528 < T < 6.792$ cm, chosen $T = 5$ cm
Total Height H	$H = T + F$	$H = 7.5$ cm

Hull of the boat




Bateau

Criteria:

1. **Waterproofing**
   Measures resistance to water and humidity, essential for environments exposed to water.
2. **Weight**
   Affects performance and efficiency, especially important in automotive and aerospace applications.
3. **Cost**
   Determines the material’s cost-effectiveness for large-scale production.
4. **Corrosion Resistance**
   Indicates the ability to resist oxidation and exposure to harsh chemical environments.
5. **Recyclability**
   Evaluates the material's ability to be reused, crucial for environmental sustainability.
6. **Torsion Resistance**
   Measures resistance to twisting forces without deformation.
7. **Flexural Strength**
   Evaluates the ability to resist forces that attempt to bend the material.
8. **Thermal Insulation**
   Measures the ability to reduce heat transfer, important for thermal management.
9. **Electrical Insulation**
   Indicates the material's ability to insulate against electricity, essential for safety in electrical applications.

MME-MAT DOMAIN

Decision matrix :

Material	Waterproofing	Weight	Cost	Corrosion Resistance	Recyclability	Torsion Resistance	Flexural Strength	Thermal Insulation	Electrical Insulation
Carbon Fiber	Medium	Very Light	Very High	Excellent	Low	Excellent	Excellent	Low	Low
ABS	Good	Heavy	Low	Medium	Good	Medium	Medium	Good	Good
Fiberglass	Good	Light	Medium	Excellent	Good	Good	Very Good	Medium	Medium

ESE DOMAIN

1. Objectives of the domain
2. Buoyancy principle
3. Notions of Stability
4. Application calculations
5. Influencing factors (Fluid Dynamics and Forces)
6. Environmental Impact



Animator: NJOUGUEP Emmanuelle

❖ Objectives of the domain

The objective of this document is to study the key concepts of buoyancy and stability applied to a mini-boat with the following specifications:

Length (L): 70 cm = 0.7 m,

Maximum width (beam): 16.98 cm = 0.1698 m,

Mass: 3.5 kg,

Weight (P): 3.5 x 9.81 = 34.335

We will determine the boat's ability to float and remain stable through calculations related to buoyancy, initial stability (metacentric height), and the righting moment.

❖ Buoyancy principle

Buoyancy is the ability of a boat to stay afloat due to Archimedes' principle. This force is equal to the weight of the volume of water displaced by the boat.

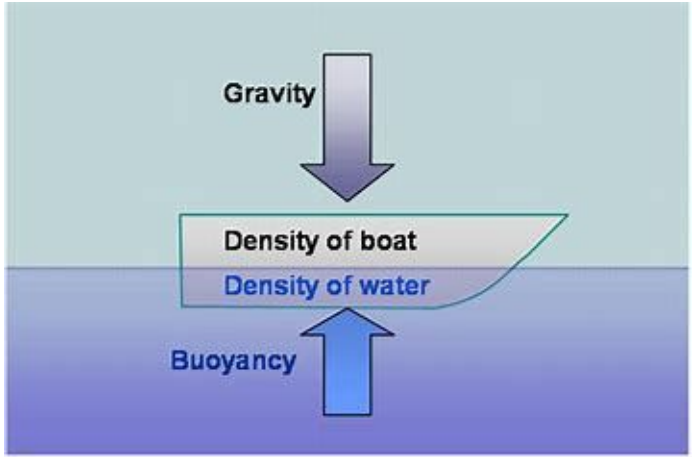
I - Archimedes' Principle :

The buoyant force (Fa) is given by : $Fa = \rho.g.v$

ρ : Density of fresh water (kg / m^3),

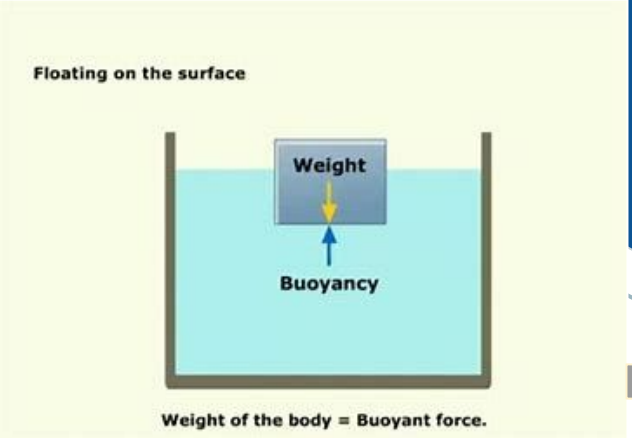
g : Gravitational acceleration (m / s^2),

v : Volume of water displaced by the hull (m^3).



II- Flotation conditions of a boat :

- If Fa > P : The boat will rise further out of the water.
- If Fa < P : The boat will sink.
- If Fa = P : The boat is in flotation equilibrium.



III- Displaced Volume at Equilibrium :

At equilibrium, the boat's weight (P) equals the buoyant force : P = Fa therefore , P = ρ.v.g
So,

$$v = \frac{P}{\rho.g} \quad \underline{\text{AN}}: \quad v = \frac{34.335}{1000*9,81} \approx 0.0035m^3$$

IV- Buoyancy ratio :

The ratio between the submerged volume and the total volume of the boat (Vt): $Rf = \frac{v}{V_t}$

With Vt = 240.

V- Immersion Height of the hull (T) :

For a this boat with width and length , the displaced volume is given by : V = ½(L) x B x h
So,

$$T = \frac{2v}{B * L} \qquad \text{AN :} \qquad T = \frac{2 * 0.0035}{0.1698 * 0.7} \approx 0.059m \text{ soit } \underline{5.9 \text{ cm}}$$

➢ Centre de carême CB (h/2)

It is the center of volume (and also the center of gravity of the fluid) displaced by the hull of a ship. It is the point of application of the Archimedes thrust which makes the ship float ; CB = 5.9 / 3 = 1.97 cm above the base.

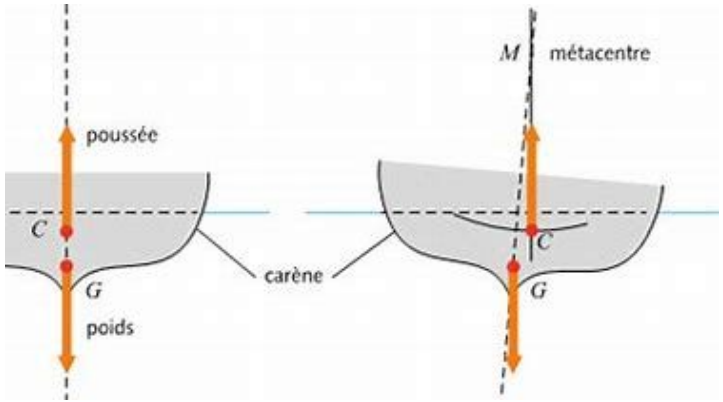
❖ Notions of stability

Stability refers to the boat's ability to resist inclinations. It is characterized by the center of gravity (G), the center of buoyancy (C), and the metacenter (M).

I- Metacentric Height (GM)

The metacentric height is given by: $GM = \frac{B^2}{12 * T}$

AN : $GM = \frac{(0.1698)^2}{12 * 0.059} \approx 0.0407m$



II- Righting Moment (Mr)

This is also called the righting torque, it is equal to the product of the boat's displacement by the horizontal distance between the verticals passing through the CG and the CC. $M_r = P * GM * \sin(\theta)$

Let's say an angle of inclination of $\theta = 12.5$; AN : $M_r = 34.335 * 0.0407 * \sin(12.5) \approx 0.303 Nm$

❖ Influencing factors

There are no factors that can influence not only the balance but also the flotation of a boat on the water, among which:

- Submerged volume (v):

The more the boat sinks, the more the volume of water displaced increases, and therefore the Archimedes' thrust.

- Hull shape:

A well-chosen and well-designed hull shape maximizes the submerged volume for a given weight, thus improving buoyancy.

- Fluid density

- Higher density improves buoyancy and promotes stability.
- Lower density requires adjustment of the submerged volume, which can compromise stability if the boat design is not adapted.

- Metacentric height (GM):

- Too large a GM causes rapid and uncomfortable rolling.
- Too small a GM can cause instability.

However, we can also note as factors, the aerodynamic and hydrodynamic forces exerted on the boat.

➔ The hydrodynamic force generated by the float

This force is the result of two forces:

- The drag force which depends on the resistance to the advancement of the float of the rudder and the fin.

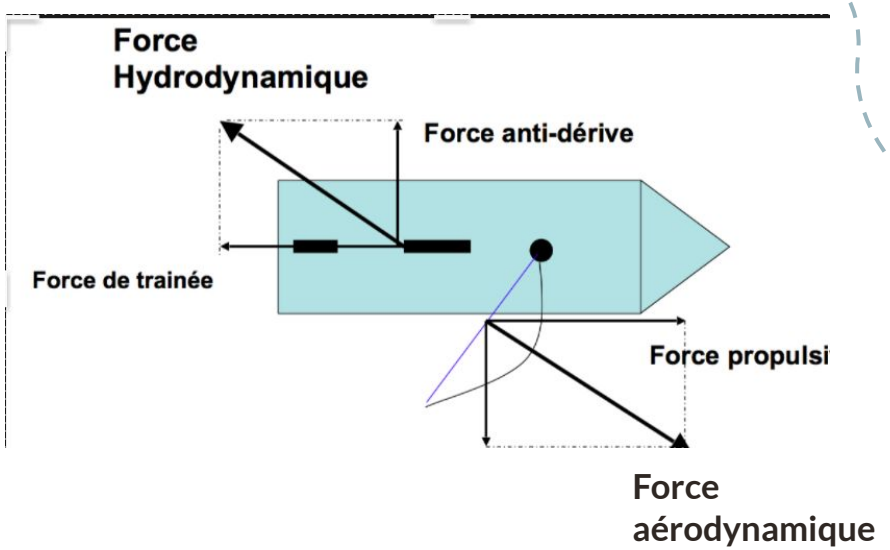
- The anti-drift force which depends on the surface and the shape of the rudder and the fin.

NB :

When the aerodynamic force is greater than the hydrodynamic force, the board accelerates.

When the aerodynamic force is less than the hydrodynamic force, the board slows down.

When the two forces are equal, the board maintains a constant speed.



❖ Environmental impact

1. Construction Materials:

Non-biodegradable materials like plastics or metals such as fiberglass for our specific case can pollute water (microplastics, corrosion).
Using eco-friendly materials, such as treated wood or biodegradable polymers, reduces environmental impact.

2. Energy Used:

Thermal engines can release fuel or oil into the water, causing chemical pollution.
Electric or solar-powered systems are cleaner, but batteries must be recycled to avoid long-term damage.

3. Impact on Flora and Fauna:

In shallow areas, the boat may disturb fish habitats and aquatic plants.
Vibrations and noise affect sensitive species, disrupting their natural behavior.

4. Generated Waste:

Activities like picnicking or fishing may lead to waste in the water.
Proper waste management is essential to preserve the ecosystem.



5. Carbon Footprint:

Manufacturing materials (plastic, metal) and transportation contribute to greenhouse gas emissions.

➢ Mitigation Measures:

- Use recyclable or biodegradable materials.

- Opt for clean propulsion systems (solar, electric).

- Limit navigation in environmentally sensitive areas.

- Always collect waste generated by users.

- Prefer local manufacturing to reduce CO2 emissions.

With these actions, the environmental impact can be significantly minimized.

EEE DOMAIN

1. Objectives of the domain
2. Notions of DC motors
3. Advantages and disadvantages of DC motor
4. Notions on the electronic components in the circuit
5. Block functional diagram of the circuit
6. Presenting the simulation with the code

Animator: Abo William Junior

EEE DOMAIN

OBJECTIVES:

The electrical and electronic engineering (EEE) section of a boat project typically covers the design, installation, and integration of all the electrical systems and electronic components on the vessel.

During the first semester, as part of the electrotechnical boat project, electronic and automatic (EEE), our mission was to design and produce a motor control circuit. This circuit had to meet two main requirements:

-Controlling the **speed** of the motor and the **direction** of rotation

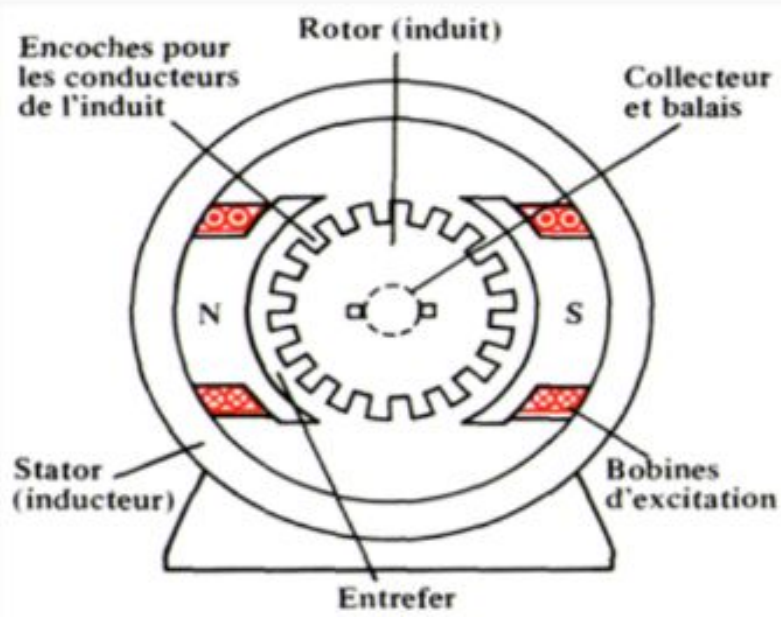
Let's first discuss on the Dc motor,

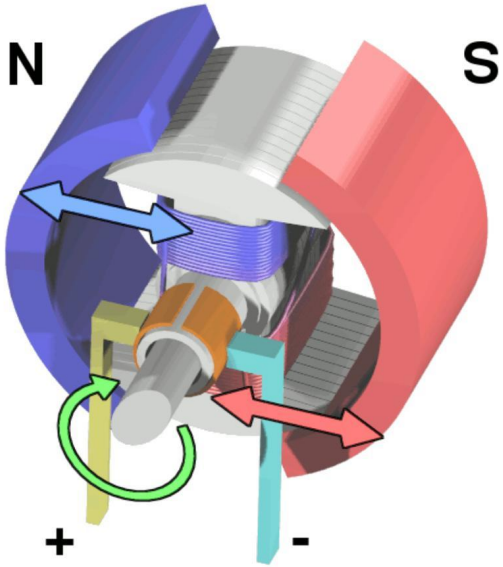
A DC motor is an electrical motor that uses direct current (DC) to produce mechanical force.The most common types rely on magnetic forces produced by currents in the coils. Nearly all types of DC motors have some internal mechanism, either electromechanical or electronic, to periodically change the direction of current in part of the motor.

ATTRIBUT	VALUE
Supply voltage	12 Vcc
Power range	36.88 W
Output speed	4289 tr/min
Shaft diameter	6.35 mm
Maximum output torque	82.08 gcm
Length	69mm
Dimensions	(∅)51.8*69 mm
Current	5.28 A
Core material	Steel Core

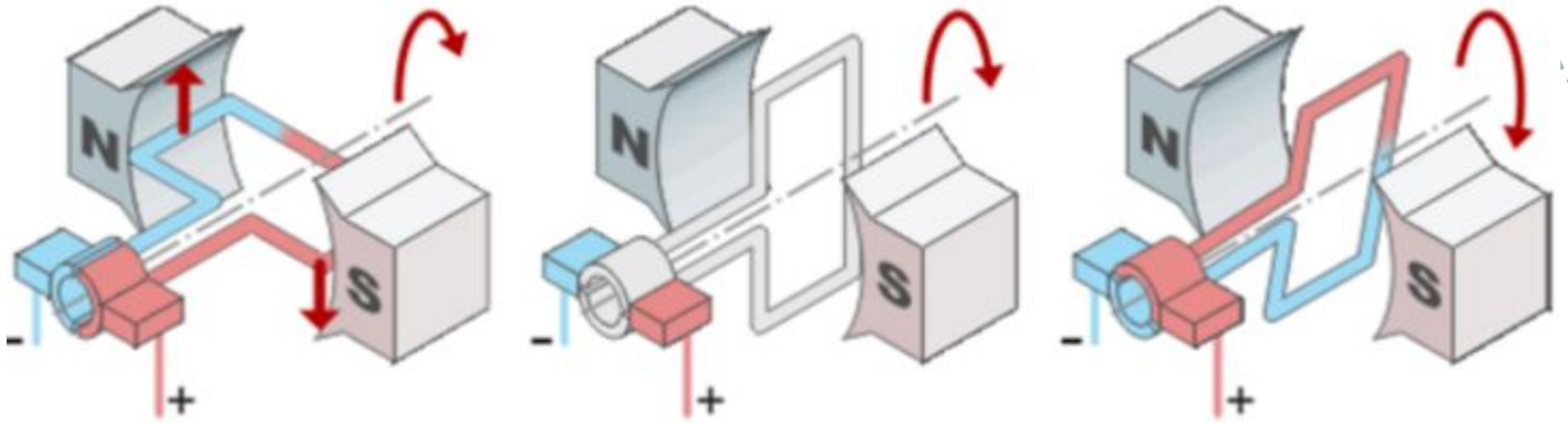
Principle of Dc motor:

A DC motor is made up of a magnetic circuit featuring a fixed part named stator, a rotative part named rotor and air-gap between both parts:The inductor (stator) creates fields which rotate by winding.An induced electrical circuit (rotor) is affected by this magnetic field. The collector and brushes provide access to the rotor circuit.

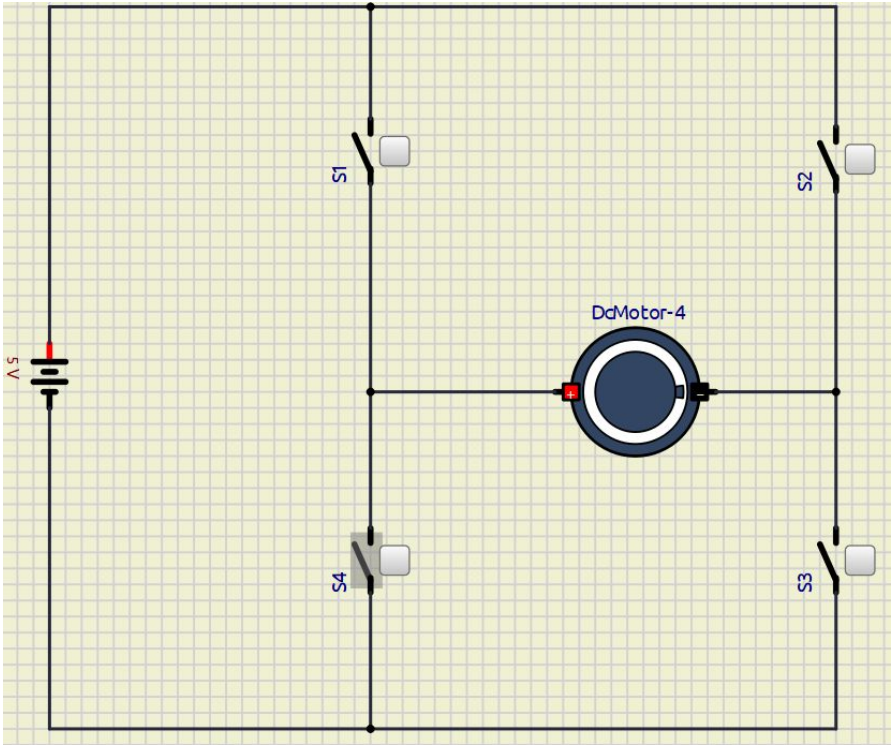


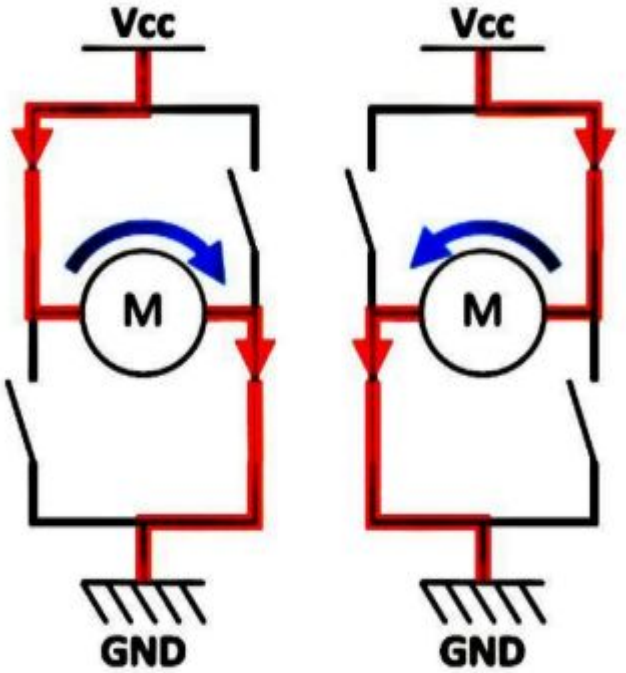


When the winding of a motor's inductor is powered by direct current, following the same principle as a permanent magnet motor (as shown in the figure below), it creates a North-South directional magnetic field (excitation flux). A coil capable of rotating about an axis of rotation is placed in the magnetic field. Additionally, the two conductors forming the coil are each electrically connected to a half-commutator and supplied with direct current via two brush contacts. According to Laplace's law (any conductor carrying a current and placed in a magnetic field is subject to a force), the conductors of the armature placed on either side of the brush axis (neutral line) are subject to equal but opposite forces F, producing a motor torque: the armature starts to rotate.



Controlling a motor to turn in 2 directions using switches, We observe that when S1 and S3 are closed(connected) while S2 and S4 are opened the motor turns towards the right direction and vice versa. It just function as an H-Bridge





Advantages of dc motor

1. DC motors have simpler installation and maintenance.
2. DC motors are typically more efficient and make better use of their input energy.
3. They have high startup power
4. They have high starting torque; it’s as high as 500% compared to normal operating torque.
5. They have fast response times to starting, stopping, and acceleration.

Disadvantages of dc motors

1. High initial cost.
2. Its operation cost and maintenance cost is very high due to the presence of commutator and brush gear.

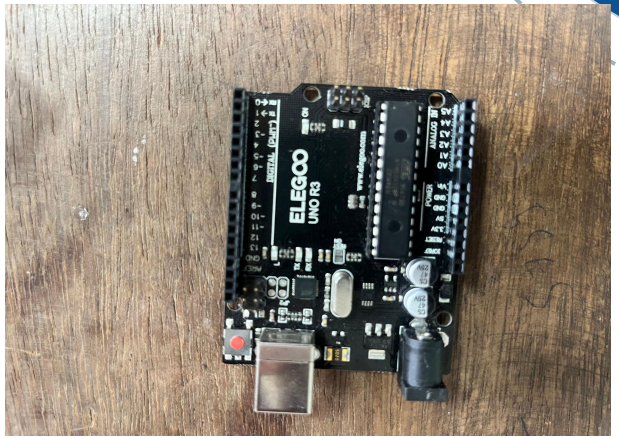


Other components present in the EEE Domaine

1-**Arduino microcontroller board**

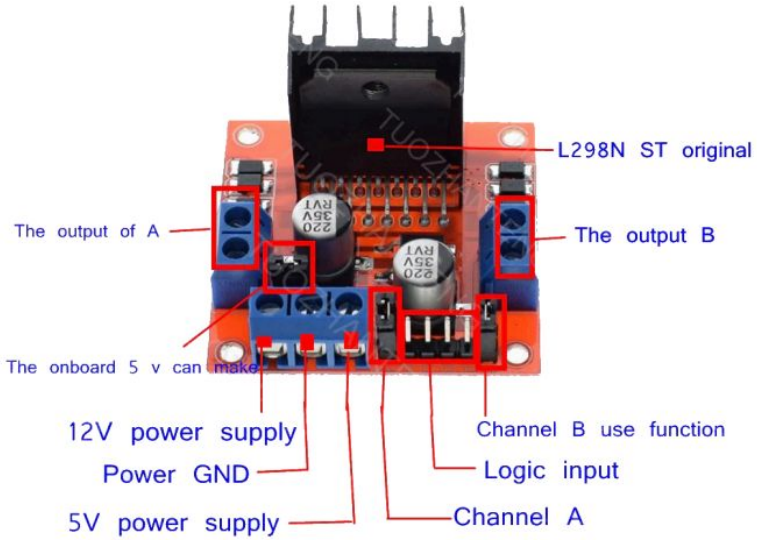
Arduino senses the environment by receiving input from many sensors, and affects its surroundings by controlling lights, motors, and other actuators.

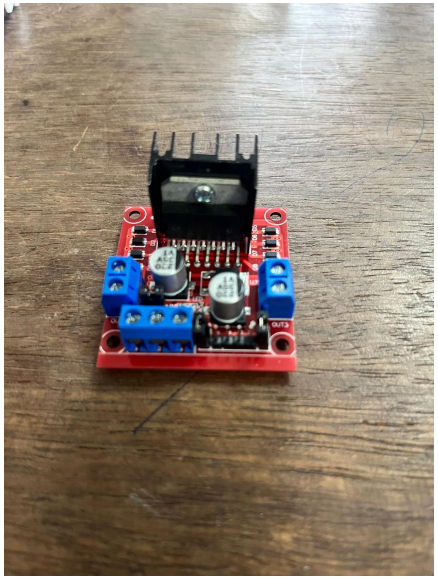




2-**DC Motor Driver(H-Bridge)**

The Dual DC Motor Driver 4A V3 allows two DC motors to be driven independently, controlling both speed and direction. The H-bridge receives the PWM signal as input and regulates the quantity energy supplied to the motor.





3-**Power**

It is essential to have an adequate source of energy to provide the voltage and current required for the operation of the H-bridge and the motor. This source can be a battery, a regulated power supply or any other source suitable energy.

4-**Button**

We've included a button to change the direction of rotation of our DC motor

5-**Potentiometer**

A user-adjustable 3 terminal variable resistor that can be used as a voltage divider. The first one will be used as a dimmer for the LCD screen, the second one will be used to set the boat direction.



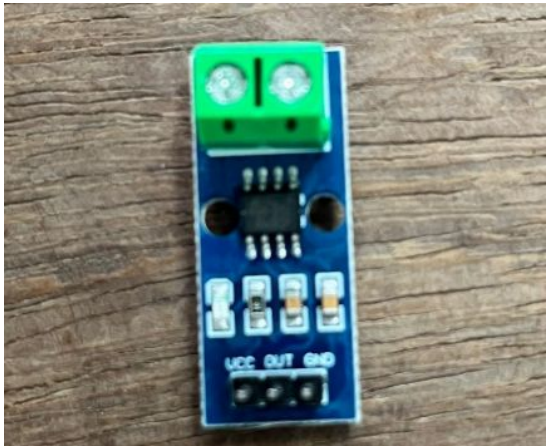


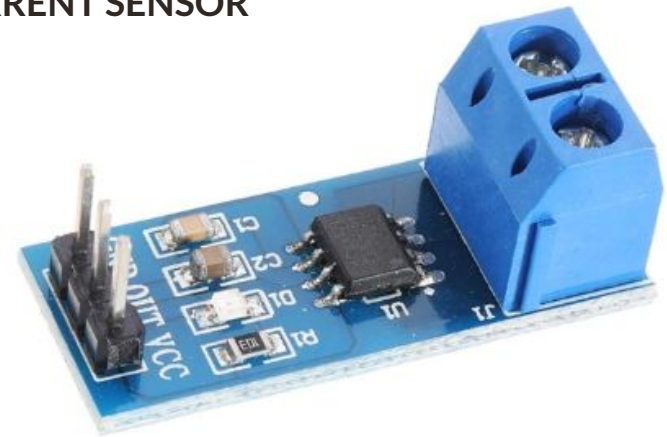
6-**Current and Speed Sensors**

Speed sensor measures the rotational speed of the motor by detecting the movement of the motor shaft and generating a corresponding output signal, allowing the speed to be calculated.

Current sensor using a sensor effect Hall, this device measures the current consumed by the motor. This information is essential for monitoring power consumption and preventing overcurrent.

CURRENT SENSOR

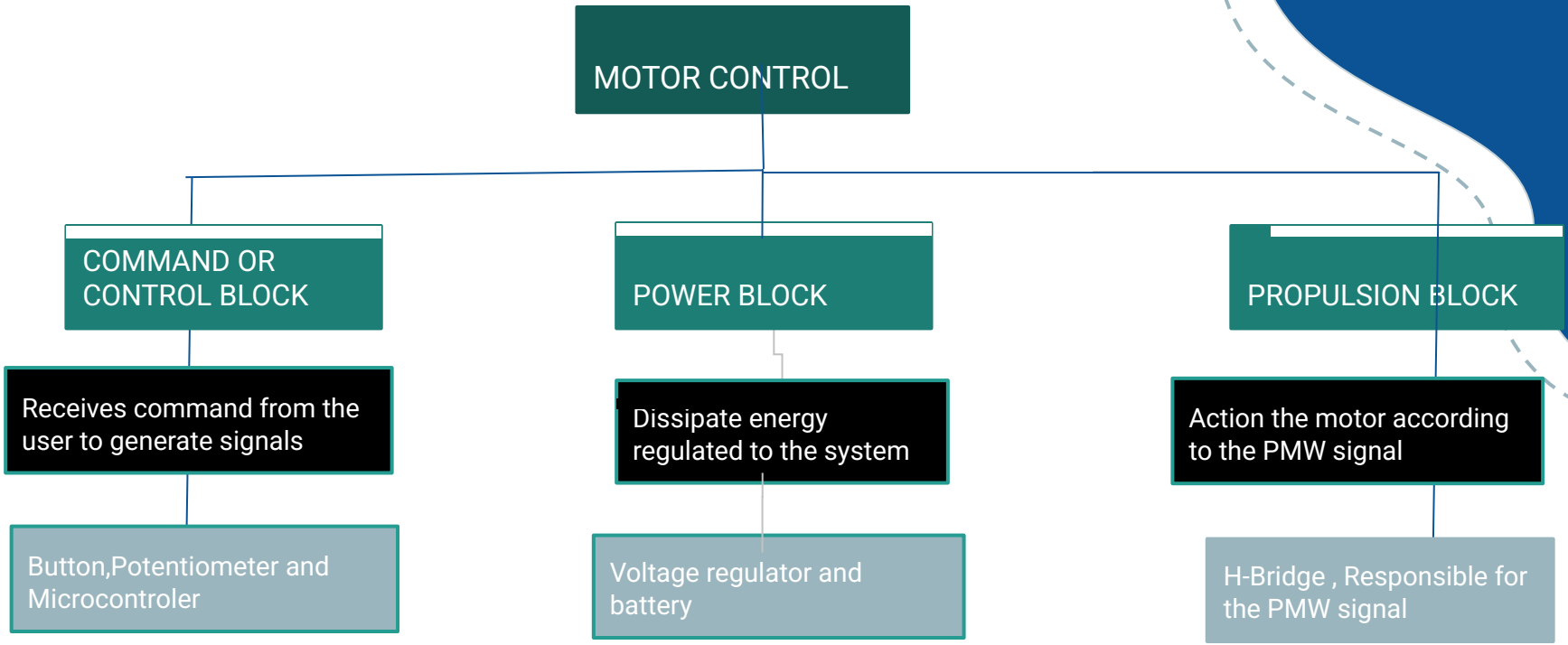




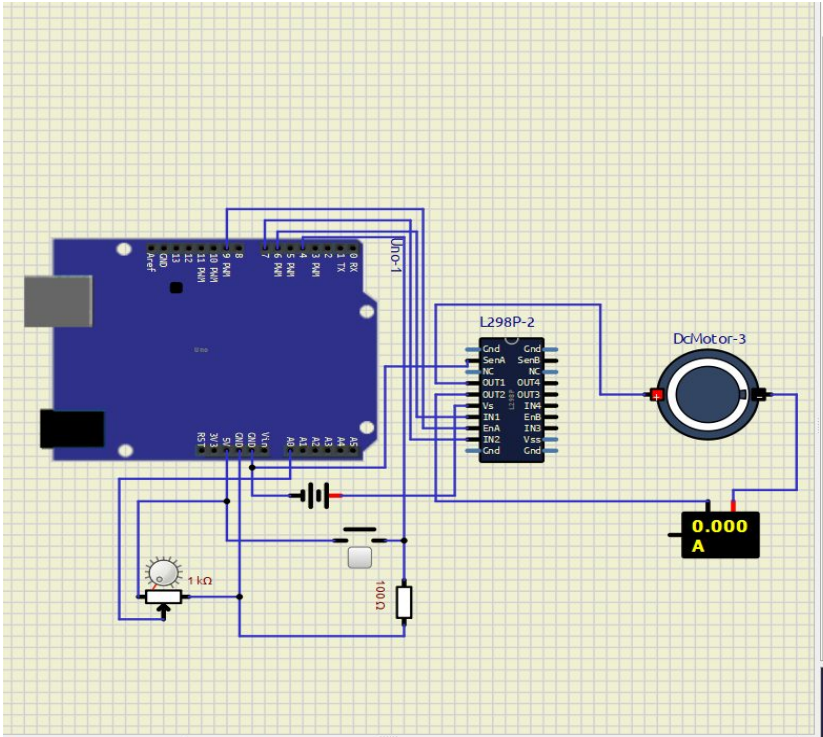
SPEED SENSOR



BLOCK FUNCTIONAL DIAGRAM OF THE CIRCUIT



UNDERSTANDING THE SIMULATION AND CODE




```
Test.ino
/*  Arduino DC Motor Control - PWM | H-Bridge | L298N  -  Example 01

    by Dejan Nedelkovski, www.HowToMechatronics.com
*/

#define enA 9
#define in1 6
#define in2 7
#define button 4

int rotDirection = 0;
int pressed = false;

void setup() {
  pinMode(enA, OUTPUT);
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
  pinMode(button, INPUT);
  // Set initial rotation direction
  digitalWrite(in1, LOW);
  digitalWrite(in2, HIGH);
}

void loop() {
  int potValue = analogRead(A0); // Read potentiometer value
  int pwmOutput = map(potValue, 0, 1023, 0 , 255); // Map the potentiometer value from 0 to 255
  analogWrite(enA, pwmOutput); // Send PWM signal to L298N Enable pin

  // Read button - Debounce
  if (digitalRead(button) == true) {
    pressed = !pressed;
  }
  while (digitalRead(button) == true);
  delay(20);

  // If button is pressed - change rotation direction
  if (pressed == true  & rotDirection == 0) {
    digitalWrite(in1, HIGH);
    digitalWrite(in2, LOW);
```

```
ulation Time:
0:00 s  000 ms  000 µs
et Speed:   100.00 %
 Speed:     000.00 %
ne Load:    000.00 %
ate Load:   000.00 %

Loading hex file:
 C:/Users/Junior/Documents/Test/Test.hex

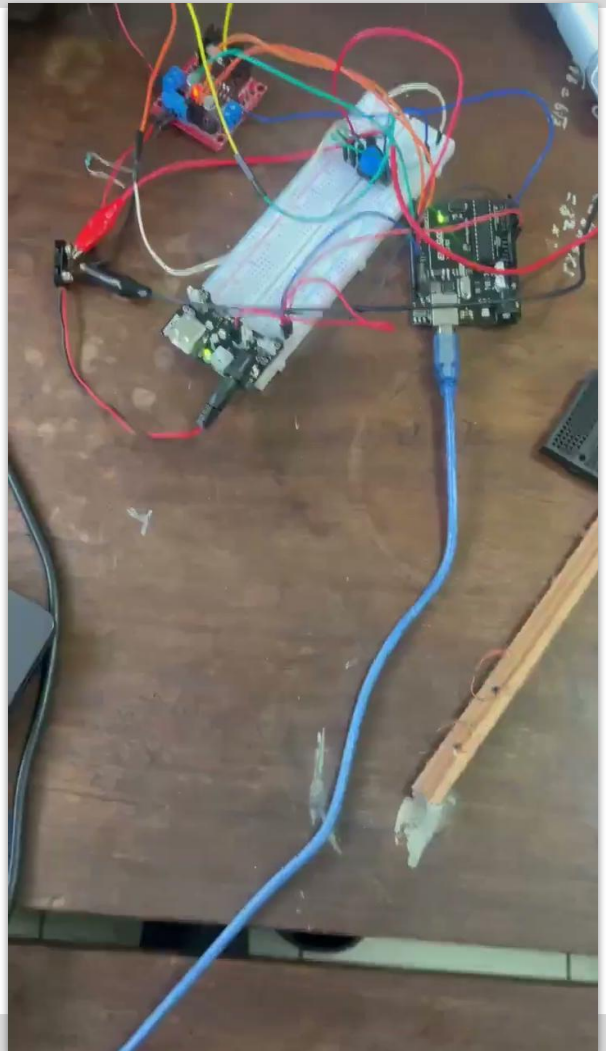
Firmware successfully loaded

Circuit Loaded:
C:\\Users\\Junior\\Documents\\Test\\Test.sim1
```

```
------------------------------------------------------------
 Fichier : C:/Users/Junior/Documents/Test/Test.ino

Found Arduino Version 2
Arduino Compiler successfully loaded.

------------------------------------------------------------
```



MIA DOMAIN

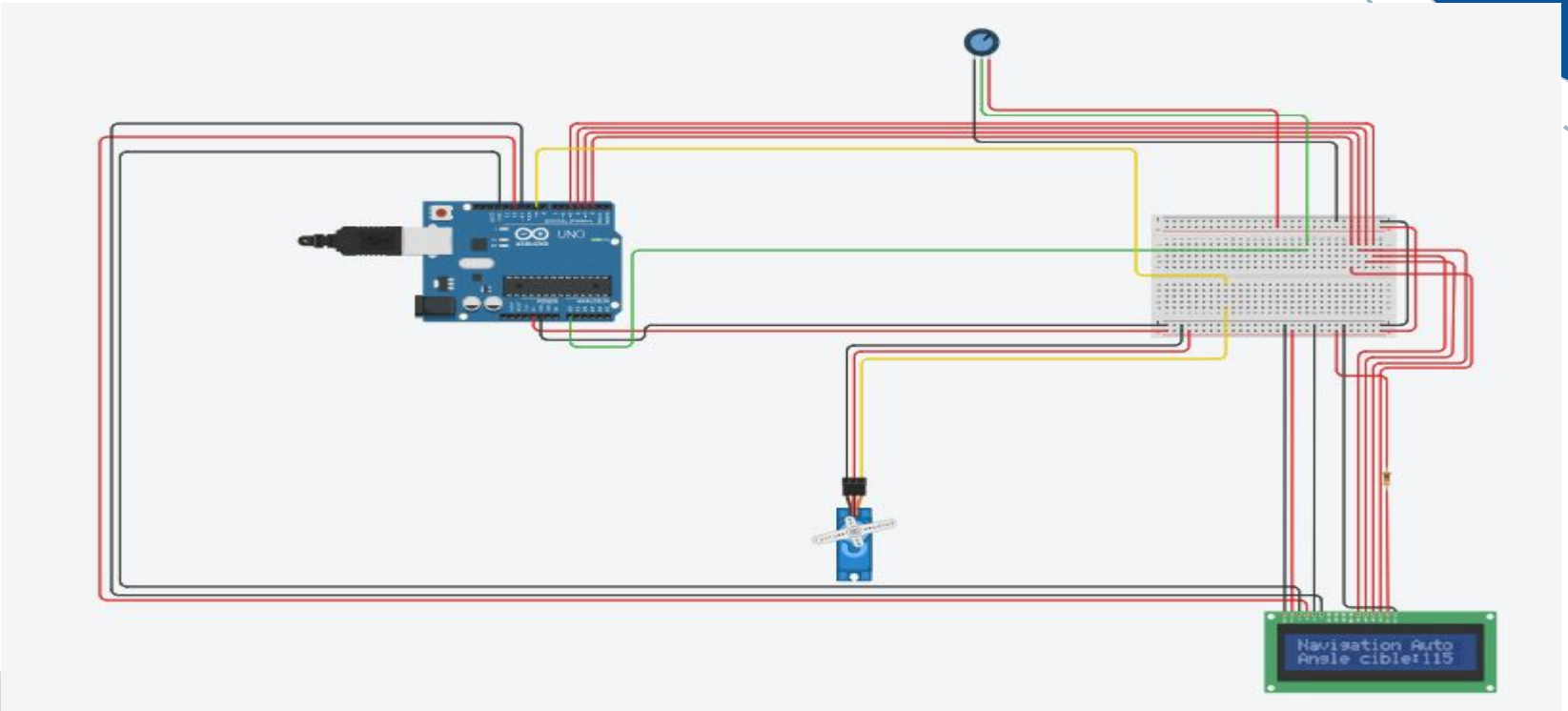
1. Objectif of the domain
2. Explanation of the model
3. Arduino code explanation

Animator: MAKOSSO Mignon

Objectif:

In this first semester the goal is to design and implement an autopilot system for a model boat, capable of maintaining the desired direction set by a potentiometer, despite external disturbances. The system includes real-time angle monitoring on an LCD for precision and control.

explanation of the model



ARDUINO CODE:

```
#include <LiquidCrystal.h>
#include <Servo.h>
#include <SoftwareSerial.h>
#include <TinyGPS++.h>

// Initialisation des composants
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
Servo myServo;
SoftwareSerial gps1Serial(10, 13), gps2Serial(8, 6);
TinyGPSPlus gps1, gps2;

// Variables
int potPin = A1, potValue, angle, currentAngle, tolerance = 5;
String lat1 = "N/A", lon1 = "N/A", lat2 = "N/A", lon2 = "N/A";

void setup() {
 lcd.begin(16, 2); lcd.print("Navigation Auto");
 myServo.attach(9);
 gps1Serial.begin(9600); gps2Serial.begin(9600);
}
```

```
void loop() {
 // Lecture du potentiomètre et ajustement de l'angle
 potValue = analogRead(potPin);
 angle = map(potValue, 0, 1023, 0, 180);
 currentAngle = myServo.read();
 if (abs(currentAngle - angle) > tolerance) {
  myServo.write(currentAngle < angle ? currentAngle + 1 : currentAngle - 1);
 }

 // Mise à jour et affichage des coordonnées GPS
 updateGPS();
 lcd.setCursor(0, 0); lcd.print("Angle: "); lcd.print(angle);
 lcd.setCursor(0, 1); lcd.print("Lat1: "); lcd.print(lat1); lcd.print(" Lon1: "); lcd.print(lon1);
}

void updateGPS() {
 while (gps1Serial.available() > 0) gps1.encode(gps1Serial.read());
 while (gps2Serial.available() > 0) gps2.encode(gps2Serial.read());
 if (gps1.location.isUpdated()) {
  lat1 = String(gps1.location.lat(), 6); lon1 = String(gps1.location.lng(), 6);
 }
```

Test from circuit

```
if (gps2.location.isUpdated()) {
   lat2 = String(gps2.location.lat(), 6); lon2 = String(gps2.location.lng(), 6);
  }
}
```

CONCLUSION



Thanks for Your Attention!